

Reliable Long-Horizon Forecasting of Terminal Voltage and Coupled Responses in Vanadium Redox Flow Batteries Under Varying Conditions

Supplementary Materials

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This supplementary document provides additional implementation and evaluation details that support the main paper.

What Table I clarifies:

- The same training protocol and hyperparameter search space were applied to all learning-based baselines to avoid tuning bias.

What Table II clarifies:

- The dataset-observable reduced-form terms used in the supplementary residual analysis are constructed from measured channels only; no unmeasured quantities are introduced.
- The residual interpretation remains subject to the observability constraints of each dataset and reduces to an electrical-side check when hydraulic or auxiliary terms are unavailable.

What Table III clarifies:

- Both supplementary metrics are computed from the same normalized balance residual used for the supplementary analysis.
- r_τ captures the frequency of large residual excursions, whereas NRMSE_ε summarizes the overall residual magnitude, providing complementary support to the main-paper physics-consistency results.

What Table IV clarifies:

- The cross-condition protocol for Table VI in the main paper is fixed across all compared variants.
- Physics-model mismatch is introduced only at test time, with network weights unchanged.
- The reported differences therefore reflect robustness under the specified protocol rather than changes in data splitting or retraining.

Clarification of C_{phys} Evaluation Scope: The self-built-platform value in Table II was computed over the dataset-level evaluation window used for multi-dataset consistency summarization. By contrast, the values in Tables V–VI were obtained under the cross-condition protocol specified in Table VII. The

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TABLE I
REPRODUCIBILITY SETTINGS. DATA SPLIT, WINDOWING, AND EVALUATION FOLLOW THE MAIN PAPER.

Item	Setting
Implementation	PyTorch (single-GPU training)
Backbone	Recurrent backbones with matched hidden size across learning-based baselines
Optimizer	Adam ($\beta_1 = 0.9, \beta_2 = 0.999$)
Learning rate	tuned on validation within $\{1e-4, 5e-4, 1e-3\}$
Batch size	tuned within $\{32, 64, 128\}$ under the same protocol
Epochs	early stopping on validation (patience = 10; max = 200)
Gradient clipping	global norm = 1.0
Random seeds	As reported in Table VII of the main paper
Hardware	single NVIDIA GPU (24GB class) + CPU pre-processing

Note. The same hyperparameter search space was applied to all compared learning-based baselines.

TABLE II
DATASET-OBSERVABLE REDUCED-FORM TERMS USED IN THE SUPPLEMENTARY NORMALIZED BALANCE RESIDUAL ANALYSIS.

Dataset	Observable channels used	Residual scope	Reduced-form terms
Sandia (USA)	$\{V, I\}$	Electrical only	$P_{\text{in}} = VI$, P_{out} omitted when the corresponding term is unobserved
Glazkov (Russia)	$\{V, I\}$	Electrical only	$P_{\text{in}} = VI$, P_{out} omitted when the corresponding term is unobserved
Self-built (CN)	$\{V, I, \Delta p, Q_{\text{flow}}, T_{\text{amb}}\}^a$	Electrical + hydraulic	$P_{\text{in}} = VI$ $P_{\text{out}} = \Delta p Q_{\text{flow}}$

Note. P_{ref} is the rated-power normalization scale used in the supplementary normalized residual analysis; in our implementation it is set to $\text{median}(|P_{\text{in}}|)$ on the training split and fixed thereafter for stable normalization. ^a T_{amb} is provided as a boundary input (Table IV in the main paper) and is not used to form P_{out} .

same C_{phys} definition was used in both cases, but the reported values correspond to different evaluation scopes and should not be read as repeated estimates from the same test segment.

TABLE III
ADDITIONAL SUPPLEMENTARY METRICS DERIVED FROM THE
NORMALIZED BALANCE RESIDUAL.

Metric	Definition (computed over rollout steps)
Violation rate r_τ	Fraction of steps with large residual excursions: $r_\tau = \frac{1}{T} \sum_{t=1}^T \mathbb{I}(\tilde{\epsilon}_{\text{bal}}(t) > \tau)$
NRMSE NRMSE_ϵ	Overall residual magnitude: $\text{NRMSE}_\epsilon = \sqrt{\frac{1}{T} \sum_{t=1}^T \tilde{\epsilon}_{\text{bal}}(t)^2}$

Note. Both metrics use the same normalized balance-residual definition in the supplementary analysis. τ is fixed to 3 across all datasets and models.

TABLE IV
EVALUATION PROTOCOL SUMMARY FOR TABLE VI.

Item	Setting
Data split	Cross-condition: $\mathcal{D}_{\text{train}}$ = low-flow , $\mathcal{D}_{\text{test}}$ = high-flow; Train/Val = 6:2 within $\mathcal{D}_{\text{train}}$; $\mathcal{D}_{\text{test}}$ is held out for evaluation.
Sampling rate	$f_s = 1$ Hz.
Sliding window / horizon	Input window $W = 60$ steps, prediction horizon $H = 60$ steps.
Long-horizon length	$H_{\text{long}} = 120$ steps.
Test-time physics-model mismatch perturbation	(1) $p = 0.10$, $\epsilon \sim \mathcal{U}[-0.10, 0.10]$; sampled once per test run and fixed over the entire prediction horizon; (2) Network weights unchanged.
Seeds	All results are reported as mean $\pm$ std over $N = 3$ random seeds: $\{0, 42, 123\}$